

Power BI Dashboard Documentation

Business Intelligence · KPI Reporting · Executive Analytics

VERSION
1.0

DATE
2025

PLATFORM
Power BI + PostgreSQL

TEAM
Data Engineering

01

Executive
DASHBOARD

02

Sales
DASHBOARD

03

Marketing
DASHBOARD

04

Customer
DASHBOARD

TABLE OF CONTENTS

Document Index

Section	Title	Pg
00	Project Overview & Architecture	3
01	Executive Performance Dashboard	4
02	Sales Performance Dashboard	5
03	Marketing Campaign Dashboard	6
04	Customer Experience Dashboard	7
05	Overall Business Impact	8
06	Technology Stack Summary	8

SECTION 00

Project Overview

Platform purpose and data architecture summary

The AI-Powered Real Estate Decision Intelligence Platform uses Power BI to transform operational data into actionable business insights. The dashboard suite serves executives, sales managers, marketing teams, and business analysts to monitor performance, identify trends, and make data-driven decisions. Dashboards connect to a PostgreSQL data warehouse fed by a Python ETL pipeline, with automated KPI monitoring via n8n workflows and Gemini AI natural language query support.

DASHBOARD 01

Executive Performance

High-level company performance for senior management

Overview & Purpose

The Executive Dashboard provides a high-level overview of company performance and serves as the primary decision-making dashboard for senior management. It consolidates all critical KPIs into a single real-time view, enabling faster strategic responses.

Key Performance Indicators

\$ Total Revenue Aggregated revenue across all regions and periods	# Sales Transactions Total count of completed property deals	% Lead Conversion Rate Ratio of qualified leads converted to sales
* Customer Rating Average satisfaction score across all transactions	~ Active Properties Number of listings currently on the market	R Revenue by Region Geographic breakdown of revenue contribution

Business Findings

Revenue Performance Significant revenue generated across regions, with specific locations contributing disproportionately to total sales.
Conversion Analysis Lead conversion rates reflect sales effectiveness. Lower rates may signal qualification or follow-up gaps.
Customer Satisfaction Declining rating trends may indicate operational issues requiring immediate executive attention.

Strategic Recommendations

- 1

Invest in High-Performing Regions

Increase resource allocation where ROI is strongest.
- 2

Improve Lead Qualification

Implement scoring frameworks to filter high-intent leads.
- 3

Customer Feedback Programs

Launch structured feedback cycles to track satisfaction.
- 4

Executive KPI Alerts

Configure n8n to notify executives on threshold breaches.

Future Improvements

» Revenue Forecasting	» AI Executive Summaries
» Predictive Lead Scoring	» Automated Anomaly Detection

DASHBOARD 02

Sales Performance

Agent effectiveness, property transactions and regional analysis

Overview & Purpose

The Sales Dashboard empowers management to evaluate individual agent performance, monitor property transactions, and identify top-performing agents and regions. It drives accountability and enables targeted coaching by surfacing granular sales-level data.

Key Performance Indicators

\$ Total Sales Cumulative sales value across all agents	A Revenue by Agent Individual agent contribution to revenue	R Revenue by Region Regional breakdown of total sales figures
# Property Sales Count Number of properties successfully sold	V Average Deal Value Mean transaction size per closed deal	T Sales Trend Month-over-month sales volume trajectory

Business Findings

Top Performing Agents A small group of agents drives the majority of total sales volume — Pareto distribution confirmed.
Regional Opportunities Certain regions consistently outperform others, signalling stronger underlying market demand.
Seasonal Sales Trends Monthly fluctuations reveal potential seasonal demand patterns that can guide campaign timing.

Strategic Recommendations

- 1

Replicate Top-Agent Strategies
Document and systematise techniques of high performers.
- 2

Coaching for Underperformers
Set targeted development plans with tracked milestones.
- 3

Focus on High-Demand Regions
Redirect sales effort toward regions with proven demand.
- 4

Monitor Monthly Fluctuations
Build seasonal forecasting into sales planning cycles.

Future Improvements

- »

Agent Performance Forecasting
- »

Automated Target Tracking
- »

Commission Analytics
- »

Territory Optimisation

DASHBOARD 03

Marketing Campaign

Campaign ROI, lead quality and budget effectiveness analysis

Overview & Purpose

The Marketing Dashboard evaluates the effectiveness and return on investment of all marketing campaigns. It provides granular visibility into spend vs. revenue, lead quality metrics, and cost-per-acquisition data — enabling the team to continuously optimise channel allocation.

Key Performance Indicators

\$ Campaign Spend Total budget allocated per campaign	R Revenue Generated Revenue directly attributed to campaign activity	% Campaign ROI Return on investment per marketing channel
L Leads Generated Total prospects acquired per campaign	C Conversion Rate Percentage of leads that converted to sales	P Cost Per Lead Average acquisition cost per generated lead

Business Findings

Campaign Effectiveness Some campaigns produced significantly higher revenue with lower spend — indicating superior targeting.
ROI Analysis High-spend campaigns do not consistently deliver the highest returns, challenging budget assumptions.
Lead Quality vs Quantity High-volume lead campaigns may not generate qualified prospects — quality metrics must be prioritised.

Strategic Recommendations

- 1

Reallocate to High-ROI Campaigns
Shift budget toward proven high-return channels.
- 2

Eliminate Underperformers
Cut spend on channels that consistently underdeliver.
- 3

Prioritise Lead Quality
Score leads at entry point to filter low-intent prospects.
- 4

Monitor Conversion Rates
Set conversion benchmarks and track against them weekly.

Future Improvements

- »

Marketing Attribution Modelling
- »

AI Campaign Recommendations
- »

Budget Optimisation Algorithms
- »

Predictive Campaign Analysis

DASHBOARD 04

Customer Experience

Satisfaction metrics, review analysis and service quality tracking

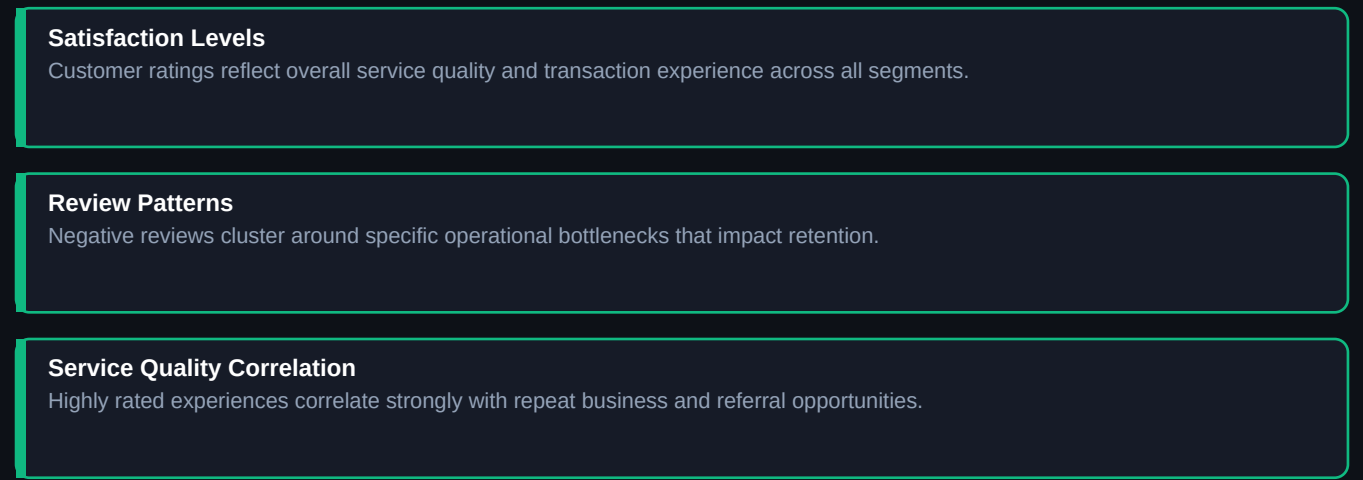
Overview & Purpose

The Customer Experience Dashboard measures satisfaction levels and service quality across all client interactions. It centralises review data, rating distributions, and feedback trends, enabling the operations team to identify pain points and proactively improve the client journey.

Key Performance Indicators



Business Findings



Strategic Recommendations

- 1

Investigate Recurring Complaints
Categorise negative reviews to identify systemic issues.
- 2

Service Improvement Initiatives
Launch targeted programs to address root-cause gaps.
- 3

Monthly Satisfaction Tracking
Report satisfaction trends at each management review.
- 4

Automated Feedback Alerts
Trigger n8n alerts when average rating drops below threshold.

Future Improvements

- »

AI Sentiment Analysis
- »

Customer Churn Prediction
- »

Automated Review Classification
- »

Customer Lifetime Value

SECTION 05

Overall Business Impact

How the Power BI ecosystem delivers measurable value

Real-Time Business Visibility

Live KPI monitoring across all departments via connected Power BI tiles.

Automated KPI Monitoring

n8n workflows trigger alerts when thresholds are breached, ensuring no metric goes unnoticed.

Faster Executive Decision-Making

Consolidated dashboards reduce time-to-insight from days to seconds.

Improved Sales Performance Tracking

Agent and regional analytics surface opportunities for coaching and resource reallocation.

Better Marketing ROI Analysis

Campaign-level attribution enables precise budget optimisation and channel pruning.

Enhanced Customer Experience

Feedback trends and satisfaction metrics drive proactive service improvements.

SECTION 06

Technology Stack

Component	Role & Responsibility
Python · Pandas	ETL pipeline — data cleaning, validation and transformation
PostgreSQL	Relational data warehouse — analytical queries, views and indexes
n8n Automation	Event-driven orchestration — alerts, scheduling and notifications
Gemini AI	Natural language query engine — NL to SQL translation
Power BI	Business intelligence layer — interactive reports and KPI tiles

The integration of Python, PostgreSQL, n8n, Gemini AI, and Power BI creates a scalable business intelligence solution capable of supporting future predictive analytics and AI-driven decision support systems.